

## PHYS 111: Tutorial 2.

- A particle moves according to the equation  $x = 10t^2$  where  $x$  is in meters and  $t$  is in seconds.
  - Find the average velocity for the time interval from 2.00 s to 3.00 s.
  - Find the average velocity for the time interval from 2.00 s to 2.10 s.
- The position of a pinewood derby car was observed at various times; the results are summarized in the following table. Find the average velocity of the car for
  - The first second
  - The last 3 s,
  - The entire period of observation.

|               |   |     |     |      |      |      |
|---------------|---|-----|-----|------|------|------|
| $t(\text{s})$ | 0 | 1.0 | 2.0 | 3.0  | 4.0  | 5.0  |
| $x(\text{m})$ | 0 | 2.3 | 9.2 | 20.7 | 36.8 | 57.5 |

- Find the instantaneous velocity of the particle described in Figure 1 at the following times:
  - $t = 1.0$  s, (b)  $t = 3.0$  s, (c)  $t = 4.5$  s, and (d)  $t = 7.5$  s.

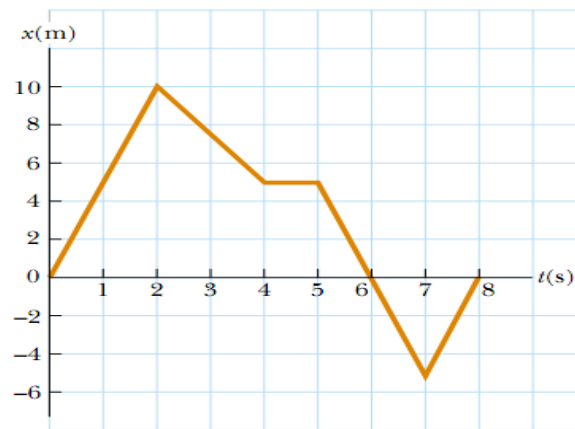


Figure 1

- A velocity–time graph for an object moving along the x-axis is shown in Figure 2.
  - Plot a graph of the acceleration versus time.
  - Determine the average acceleration of the object in the time intervals  $t = 5.00$  s to  $t = 15.0$  s and  $t = 0$  to  $t = 20.0$  s.

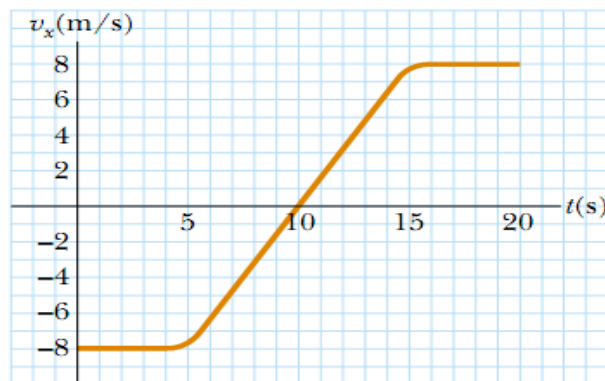


Figure 2

5. A truck covers 40.0 m in 8.50 s while smoothly slowing down to a final speed of 2.80 m/s.
  - (a) Find its original speed.
  - (b) Find its acceleration.
6. Starting from a pillar, you run 200 m east (the +x-direction) at an average speed of 5.0 m/s and then run 280 m west at an average speed of 4.0 m/s to a post. Calculate:
  - (a) Your average speed from pillar to post
  - (b) Your average velocity from pillar to post.
7. An electron in a cathode ray tube (CRT) accelerates from  $2.00 \times 10^4$  m/s to  $6.00 \times 10^6$  m/s over 1.50 cm.
  - (a) How long does the electron take to travel this 1.50 cm?
  - (b) What is its acceleration?
8. In the fastest measured tennis serve, the ball left the racquet at 73.14 m/s. A served tennis ball is typically in contact with the racquet for 30.0 ms and starts from rest. Assume constant acceleration.
  - (a) What was the ball's acceleration during this serve?
  - (b) How far did the ball travel during the serve?
9. In Mangochi, Malawi, the ultimate test of a young man's courage once was to jump off a bridge into the Shire River, 23.0 m below the bridge.
  - (a) How long did the jump last?
  - (b) How fast was the diver traveling upon impact with the water?
  - (c) If the speed of sound in air is 340 m/s, how long after the diver took off did a spectator on the bridge hear the splash?
10. A rock is thrown straight up and reaches a height of 10 m.
  - (a) How long was the rock in the air?
  - (b) What is the initial velocity of the rock?
11. A student throws a set of keys vertically upward to her sorority sister, who is in a window 4.00 m above. The second student catches the keys 1.50 s later
  - (a) With what initial velocity were the keys thrown?
  - (b) What was the velocity of the keys just before they were caught?
12. A basketball dropped from rest 1.00 m above the floor rebounds to a height of 0.67 m. Assuming the ball is not moving horizontally, calculate its velocity...
  - (a) just before it hits the floor on the way down and
  - (b) Just after it left the floor on the way up.
  - (c) If the ball is in contact with the floor for 0.10 s determine its acceleration...
    - i. On the way down,
    - ii. While it is contact with the floor, and
    - iii. On the way up.